

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 40)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 1

1

Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1

2

3

4

5

0 1

2

Explain why involving local communities is essential to the success of the reserve.

[3 marks]

0 1

3

Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

10

0 2 . 1

Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 2 . 2

Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 2 . 3

Explain why tuna aquaculture cannot yet replace tuna fishing.

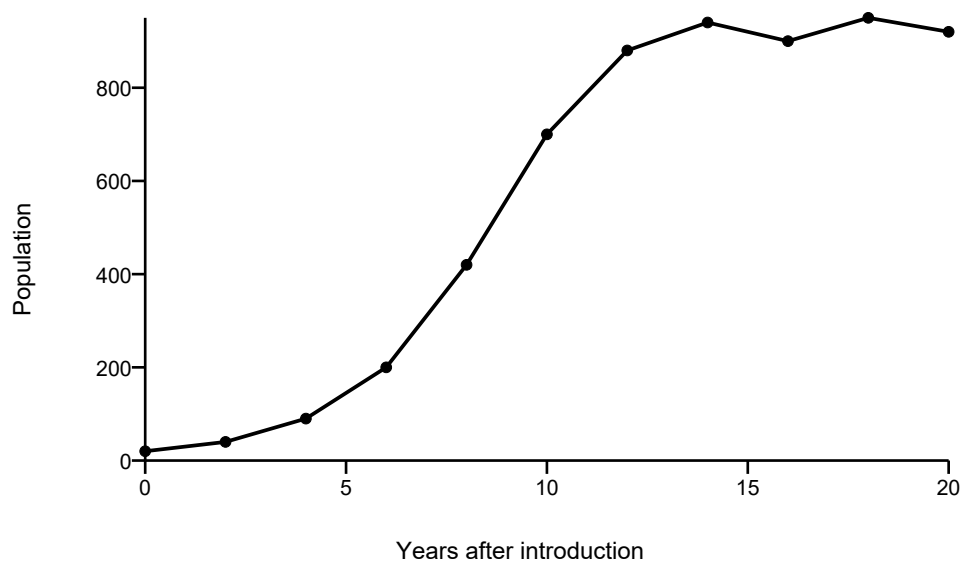
[2 marks]

10

0 3

Figure 1 shows the population of red deer on an island after they were introduced in year 0. No predators are present.

Figure 1



0 3

. 1

Name the shape of the growth curve and explain the changes in growth rate shown.

[4 marks]

03 . 2

Explain the difference between density-dependent and density-independent factors, with an example of each.

[2 marks]

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03 . 3

Explain why the deer population on the island may need to be culled even though it appears stable.

[2 marks]

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03 . 4

Explain how predator and prey populations regulate each other.

[2 marks]

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10

04 . 1

Explain why tropical rainforests have such high biodiversity.

[4 marks]

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0 4 . 2

Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

0 4 . 3

Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

0 4 . 4

Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

10

0 5 . 1

Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

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Dissolved oxygen

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pH

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0 5 . 2

Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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0 5 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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10

0 6 . 1

Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

1

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2

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0 6 . 2

Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 6 . 3

Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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5

0 7

Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

07 . 1

Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

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07 . 2

Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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07 . 3

Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

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07 . 4

Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

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10

08

A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

08

1

Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

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08

2

Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

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08

3

Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

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10

09

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

09

. 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer

thousand tonnes

09

. 2

Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer

thousand tonnes

09

. 3

Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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09 . 4 Describe how each of the following measures obtained from commercial catches can indicate overfishing. **[4 marks]**

Catch per unit fishing effort

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Mean fish size and mean age

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10

10 . 1 Discuss the strategies that can be used to increase the sustainability of agriculture. **[9 marks]**

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[illegible]

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10 . 2 State what is meant by **permaculture**.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS